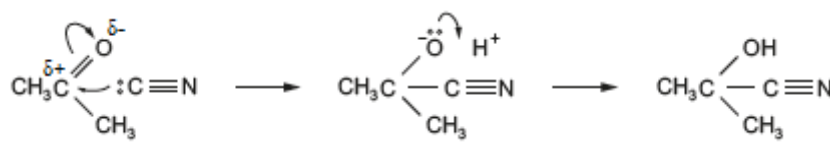
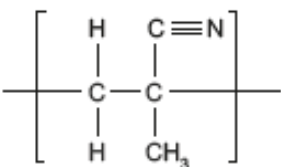


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)	(i)		 nucleophilic addition (1) curly arrows (1) partial / full charges (1)	3			3		
		(ii)		incomplete extraction by ethoxyethane (1) incomplete distillation / decomposition of product (1)			2	2		2
		(iii)		moles of propanone used = $17.4 / 58.06 = 0.300$ (1) 1:1 molar ratio $\Rightarrow$ 0.300 mol of 2-hydroxy-2-methylpropanenitrile mass of 2-hydroxy-2-methylpropanenitrile = $0.300 \times 85.07 = 25.5$ g (1) percentage yield = $18.6 \times 100 / 25.5 = 73 \%$ (1)		3		3	2	
		(iv)				1		1		